

DAY 05

Natural Language Processing Concepts

Professional Portfolio Notes • Microsoft Learn Day 05

Learner	MD Mostakim Hossain
Focus	AI-assisted text analysis for SOC workflows
Assessment	Passed • Score: 100%
Status	Completed

Portfolio Summary

Completed a foundational NLP module covering tokenization, statistical text analysis, semantic embeddings, and browser-based text analytics exercises.

Learning Objectives

- Explain how NLP enables machines to process and analyze natural-language text.
- Describe tokenization and common preprocessing techniques.
- Compare frequency analysis, TF-IDF, Bag of Words, and TextRank.
- Explain embeddings, cosine similarity, attention, and contextual meaning.
- Apply text summarization and language detection in hands-on playgrounds.
- Connect NLP capabilities to practical SOC analyst workflows.

1. NLP and Text Analysis

Natural Language Processing (NLP) is a branch of artificial intelligence that enables machines to process human language. Text analysis uses NLP to extract meaning, structure, entities, categories, sentiment, and concise summaries from unstructured text.

Capability	Purpose	Security Example
Language detection	Identify the primary language	Route a foreign-language phishing message for analysis
Key term extraction	Identify important words or phrases	Highlight urgent, verify, password, or suspicious link
Entity recognition	Identify named or structured entities	Extract IPs, domains, usernames, and organizations
PII detection	Locate sensitive personal information	Redact personal details before sharing or reporting
Classification	Assign text to a category	Classify email as legitimate, spam, or phishing
Summarization	Reduce text while preserving key points	Create a concise incident overview

2. Tokenization and Preprocessing

Tokenization breaks text into smaller units called tokens. Depending on the tokenizer, a token may be a word, a word fragment, or punctuation. Token IDs provide a consistent numerical reference for repeated tokens.

Technique	Description
Normalization	Converts text to a consistent format, such as lowercase and standardized punctuation.
Stop-word removal	Removes low-information words when doing so does not change meaning.
N-gram extraction	Analyzes adjacent token groups: unigram, bigram, or trigram.
Stemming	Uses rules to reduce words toward a common stem.
Lemmatization	Uses grammar and vocabulary to find a valid base form.
Part-of-speech tagging	Identifies grammatical roles such as noun, verb, and adjective.

Security Example

FAILED LOGIN and failed login can be normalized to the same form. The phrase failed login is a bigram; failed login attempt is a trigram.

3. Statistical Text Analysis

Frequency Analysis

Frequency analysis counts token occurrences to estimate a document's topic. Frequency alone may overvalue words that are common across many documents.

TF-IDF

Term Frequency–Inverse Document Frequency combines a term's frequency within one document with its rarity across a collection. A term that is frequent in one document but uncommon elsewhere receives greater discriminative importance.

Key Rule

Frequent in one document + rare across the corpus = higher TF-IDF.
Common across most documents = lower TF-IDF.

Bag of Words

Bag of Words represents text by token occurrence or frequency while usually ignoring grammar and word order. This makes it efficient, but the sentences 'Dog bites man' and 'Man bites dog' may receive equivalent representations despite having different meanings.

TextRank

TextRank represents sentences as nodes and their similarities as weighted edges. Highly connected sentences can be selected for extractive summarization. At word level, the same concept can support keyword extraction.

4. Semantic Language Models

Semantic models represent tokens as multidimensional vectors called embeddings. Similar meanings tend to produce similarly oriented vectors, enabling comparison with measures such as cosine similarity.

Concept	Meaning
Vector	An ordered numerical array representing learned features.
Embedding	A learned vector representation of a token, sentence, or document.
Cosine similarity	Compares vector orientation to estimate semantic similarity.
Attention	Weights contextually important tokens when interpreting meaning.
Contextual embedding	Represents the same token differently when its meaning changes by context.

Context Example

The word bank refers to a financial institution near deposited and money, but to a river edge near sat and river.

Vector Relationships

- $\text{dog} + \text{young} \approx \text{puppy}$
- $\text{cat} + \text{young} \approx \text{kitten}$
- $\text{puppy} - \text{young} \approx \text{dog}$
- $\text{Puppy} : \text{Dog} :: \text{Kitten} : \text{Cat}$
- Vector arithmetic usually returns a nearest semantic match rather than an exact guaranteed token.

5. Hands-on Exercise

The browser-based exercise used a generative chat playground and a specialized language playground to explore practical text analytics.

1. Configured an instruction directing the AI assistant to analyze and summarize text.
2. Submitted a long product review and requested a single short-paragraph summary.
3. Used a specialized language analyzer to detect the primary language of sample text.
4. Reviewed prediction confidence as an estimate rather than a guarantee.
5. Reinforced the need to validate AI-generated conclusions against source material.

6. Assessment Evidence

Knowledge Check	Verified Answer
Purpose of tokenization	Break text into smaller units for analysis
Term importance across a document collection	TF-IDF
Role of embedding vectors	Capture semantic token relationships in multiple dimensions

Result

Module assessment passed with a score of 100%.

7. SOC Analyst Applications

Workflow	NLP Contribution	Required Analyst Check
Phishing triage	Classify suspicious language and extract key phrases	Inspect sender, headers, URLs, and attachments
IOC extraction	Extract IPs, domains, users, and organizations	Verify each entity in raw evidence
Alert correlation	Group semantically similar alerts	Confirm asset, identity, and timeline context
Incident reporting	Summarize long investigations	Validate every claim against logs and timestamps
Data protection	Detect and redact PII	Confirm that sensitive fields are fully removed

Responsible Use

AI supports analysis; it does not replace evidence-based judgment. A SOC analyst remains accountable for validation, documentation, severity, and escalation decisions.

Key Takeaways

- Tokenization converts text into manageable analysis units.
- Statistical methods reveal frequency and document-specific importance.
- Embeddings represent semantic meaning in multidimensional space.
- Attention helps models interpret a token using its context.
- NLP supports summarization, keyword extraction, entity recognition, and classification.
- Security use cases require sanitization, source verification, and human oversight.

Completion

Module Component	Status
Theory units	Completed
Hands-on text analytics exercise	Completed
Module assessment	Passed — 100%
Final summary	Completed

Prepared for the AI + Cybersecurity learning portfolio of MD Mostakim Hossain.